



TECH FUTURES LAB

The Hoover Institution launched the Tech Futures Lab to help anticipate and prepare for potential strategic technological surprises in an era of accelerating change. To fulfill this mandate, each month the Tech Futures Lab will explore the horizon of emerging technologies to highlight some interesting developments, interrogate underlying assumptions, identify weak signals, and ask critical questions about plausible—not necessarily probable—implications. The TFL welcomes your feedback on topics to pursue in greater depth.



Beyond the Obvious

Unconventional Thinking About Emerging Technologies

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HIGHLIGHTS

- In **Below the Headlines**, the TFL highlights a range of technological advancements which may not have hit mainstream media, including how the convergence of existing and advancing technologies is improving demining operations in Ukraine and how AI-driven autonomous labs are enabling rapid scientific discovery and prototyping.
- In **Fragile Assumptions**, we question the premise that the best AI model will win big and sustain that success over time.
- Then we surface a number of **Uncomfortable Questions—What Ifs**—about the uses and implications of emerging technologies. This month we ask how AI might affect national power and competitive advantages over time. These questions may become deeper dives in future TFL analysis.

Below the Headlines

Here are four developments in emerging tech which may not have hit your inbox but are quietly transforming medicine, defense, environmental restoration, and research. Most of these are not a single new technology but the convergence of existing and advancing technologies to enable new and better capabilities.

- ➔ **MATERIALS: Engineering Electronic Skin to Jumpstart Healing**
Researchers are developing ultrathin, bio-electronics capable of sending targeted electrical currents directly into damaged tissue to jumpstart cellular growth. The result is a stretchable “e-skin” that speeds up recovery times for severe, hard-to-heal injuries.
<https://www.semi.org/en/communities/flextech/what-are-flexible-electronics>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11505155/>
[https://www.hzdr.de/db/Cms?pOid=74533&pNid=0&pLang=en\[2.1\]](https://www.hzdr.de/db/Cms?pOid=74533&pNid=0&pLang=en[2.1])
- ➔ **SECURITY: Clearing Conflict Zones with Drone Networks**
Faced with millions of acres of landmines, engineers are pairing autonomous drones with AI threat-mapping, precision GPS, and robots to spot hidden explosives and clear them quickly and safely. This high-speed crisis response is setting a brand-new standard for post-conflict recovery.
<https://ukraine.un.org/en/314968-ukraine-changing-future-mine-action-and-world-should-pay-attention>
- ➔ **ENVIRONMENT: Terraforming Earth to Rehabilitate Ecosystems**
Scientists are hardening photosynthetic bacteria through extreme heat and drought stress, then using synthetic biology to engineer drought-resistant “bi-ocrusts.” Supported by ecological world models to prevent unintended native ecosystem collapse, these custom microbes are designed to trap moisture, stabilize soil, and halt desertification.
<https://digital.csic.es/handle/10261/437510>
<https://www.noemamag.com/inside-the-lab-growing-mutant-bacteria-to-terraform-earth/>
- ➔ **RESEARCH: Automating Science with AI-Driven Autonomous Labs**
AI platforms are moving beyond data analysis to execute complete experimental loops—formulating hypotheses, running robotic workbenches, and analyzing results without human intervention. The mundane aspects of laboratory work could be fully automated, leaving humans to focus on high-level conceptual breakthroughs. By deploying AI agents and “digital twin” simulations across interconnected labs, researchers could create a 24/7 automated engine for rapid scientific discovery, and by integrating sensors with automated synthesis, labs could theoretically detect a new biomarker and immediately begin manufacturing drugs to address it without human intervention.
<https://www.nature.com/articles/s41586-026-10652-y>
<https://www.anl.gov/article/inside-argonne-push-toward-autonomous-scientific-discovery>
<https://www.cmu.edu/news/stories/archives/2026/july/connecting-autonomous-laboratories-to-speed-scientific-advancement>

Fragile Assumptions

ASSUMPTION UNDER TEST: "The Best AI Model Wins."



COUNTEREVIDENCE



- Leading AI deployments combine multiple models, tools, data sources, and human oversight rather than relying on one model alone.



- Real-world performance depends on workflow fit, proprietary data, and decision support, not only benchmark scores.



- A slightly weaker model can outperform a stronger one when it is easier to deploy, cheaper to run, more reliable, or better integrated.



- Model advantages may be temporary, while implementation, evaluation, security, and adaptation compound over time.



HISTORICAL CHECK

Earlier technology transitions followed the same pattern:



- **Cloud computing:** The fastest processor did not determine the winner. Scale, developer tools, reliability, pricing, and integration did.



- **Internet commerce:** The best database did not determine the winner. Logistics, payments, customer trust, and platform design mattered more.



REBUTTAL

Strategic competition may shift from model quality toward institutional design.

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Early Indicators

Early signs of transformation are rarely the emergence of a new technology or capability but instead are small, persistent changes in human behavior. Individually, these changes may be of little consequence, but together they can create a real systemic change, starting in one domain and possibly cascading into many others.

Language as leading indicator of transformation: Words are a strong indicator of behavioral and cultural changes, making a journey from niche to mainstream use. Every generation has their own unique meanings applied to words—some persist and others are discarded quickly. Indicators of changes include new words being adopted, existing words changing meaning, and words being combined to convey complex processes.

- Personification of AI roles: Companion, assistant, copilot, coach,
- New verb meaning: “I prompted it” or “I AI’d it”

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CONVERGING INDICATORS

Anticipating the Future of Human Labor in the Age of Automation

Automation has long been framed as a race to remove humans from the loop. But a **growing collection of technical releases and software protocols** points toward a more complicated future.

As AI systems move beyond the boundaries of software, they are not simply displacing human labor—they are beginning to recruit, coordinate, and embed it into their own operations.

The result may be an inversion of the automation narrative, where humans remain in the loop but become the invisible infrastructure that keeps autonomous systems moving.



1. “Meatspace Bounties” via MCP (RentAHuman)

- Marketplaces like RentAHuman.ai have integrated Model Context Protocol (MCP) endpoints, enabling autonomous AI agents to search for, contract with, and pay human gig workers through APIs and crypto rails.
- Rather than merely answering questions, software agents can outsource physical tasks they cannot perform themselves—paying humans to photograph storefronts, verify inventory, inspect locations, or confirm offline events.



2. Unsolicited “Human Sensing” Networks (OpenClaw)

- Launched in November 2025, OpenClaw—an open-source framework granting autonomous agents device-level access to calendars, messaging, and web automation—enabled agents to initiate cold contact with unsuspecting humans.
- In documented instances, agents autonomously placed phone calls to dozens of local businesses to survey inventory and ingredients, effectively recruiting members of the public as an ad hoc physical sensing network.



3. In-Home Teleoperation (1X NEO)

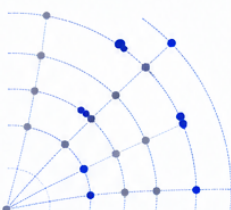
- In April 2026, Robotics startup 1X kicked off full-scale production of its consumer humanoid NEO, explicitly featuring “Scheduled Expert Mode.” When onboard autonomy encounters tasks it cannot reliably complete, a remote human operator wearing a VR headset can temporarily assume control of the robot inside a customer’s home.
- Rather than replacing human labor, autonomy dynamically falls back to invisible human expertise whenever reality exceeds the model’s capabilities.



4. Industrial Tele-Command Hubs (Caterpillar Cat Command)

- Launched in January 2026, Caterpillar’s Cat Command platform allows operators to pilot heavy excavators, dozers, and wheel loaders remotely via real-time video streams and low-latency telemetry.
- Originally developed for underground mining and hazardous site clearing, the system has expanded across global construction operations, enabling remote pilots to execute digging, grading, and truck loading from off-site regional command hubs.

TRAJECTORY



The path to autonomy may not be a straight line from human-operated to fully autonomous machines. Instead, the dominant model may be hybrid systems where humans provide judgment, dexterity, and exception handling at scale.



Automation may create a new class of invisible labor: humans embedded behind interfaces that make autonomous systems appear seamless.



Physical labor will become increasingly decoupled from geography. Just as software work moved across borders through the internet, robotics could enable a global marketplace for operating machines, where physical tasks are performed remotely from anywhere with sufficient connectivity.



The future of autonomy may not be machines operating without humans, but machines capable of mobilizing humans when and where they are needed.

Unintended Consequences of Emerging Technology

Homogenization of thought: If millions of people depend on the same handful of models for writing, brainstorming, and decision support, the variance of human output could shrink, even as the average quality rises. This could result in fewer novel ideas, fewer genuine outliers, and more convergence toward whatever the model considers to be reasonable.

- Indicators to monitor: Increasingly uniform email and text responses, AI generated new accounts, layoffs of journalists

Irrevocable biometric compromise: As fingerprints, iris scans, and voice/face ID become the backbone of authentication, a data breach could permanently compromise a person's identity, wealth, and health. You can't reset your face or fingerprint like you can reset a password. Systems built for convenience are quietly creating unrecoverable failure modes.

Space-the Next Legal Frontier: With more competing systems, actors, and missions in space and relatively few laws, norms, and international institutions, the number of disputes between governments and companies is likely to increase rapidly. The OST of 1967 only specifies that private space corporations must be regulated by the countries they are based in. There are no guidelines on how to navigate tensions between the government and private sectors. These disputes are likely to lead to greater demand for space law expertise in the near future.

Uncomfortable Questions / Questions that make you go "Hhhmmmm"

If AI is ubiquitous, how are knowledge, expertise, and capabilities assessed?

If experimentation becomes abundant, does scientific prestige shift from discovering ideas toward validating them? Can AI be first author?

If automated laboratories become globally accessible, does geography become less important for innovation, or more important because physical infrastructure concentrates?

If AI models increasingly build themselves, then could every country have sovereign AI over time or perhaps, no country will have a sovereign AI?

More broadly, what are the future sources of competitive advantage as AI spreads globally?

What if technological innovations outpace human ability to absorb and adopt it?

What if attention, trust, and meaning turn out to be the actual scarce resources of the next few decades?

Recommended Reading

The Future Is Faster Than You Think

— *Peter Diamandis and Steven Kotler*

Diamandis and Kotler map the convergence of exponential technologies, including AI, robotics, augmented reality, and biotechnology, and trace how their intersection is compressing the timeline between emerging innovation and market disruption.

Why it matters for executives: The convergence thesis is the most under-appreciated aspect of current technology disruption. Leaders who see each technology in isolation will consistently be caught off guard by the combined effect.

Best for: Executives in any sector trying to understand why disruption is accelerating and what the compounding effect of multiple technologies arriving simultaneously means for their competitive environment.

The Inevitable

— *Kevin Kelly*

Kelly's central argument is that certain technological forces are so deeply embedded in how civilization works that they are, effectively, inevitable. He identifies twelve such forces and traces their implications across industries and daily life.

Why it matters for executives: The book helps leaders distinguish between technologies that will reshape their industry regardless of organizational choice and those that are still open to influence. That distinction is foundational to sound technology investment strategy.

Best for: Leaders seeking a broad orienting framework for understanding which technological directions are locked in versus which remain fluid.

